Functional & Performance Testing

Model Performance Test

Date	21 February 2026
Team ID	LTVIP2026TMIDS80513
Project Name	TransLingua: AI-Powered Multi-Language Translator
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Text Input & Context Validation	Enter valid text and select "Academic" or "Business" context toggles.	The system accepts the text and applies the correct contextual tone.	Context applied successfully.	Pass
FT-02	Character Limit Validation	Enter text exceeding the maximum token limit for a single Gemini API call.	The system shows an error or prompts the user to shorten the text.	Warning message displayed.	Pass
FT-03	Translation Accuracy (USN-9)	Upload a document with **LaTeX formulas** and click "Translate."	Content is translated while preserving the integrity of LaTeX formulas and citations.	Formulas remain unchanged.	Pass
FT-04	API Connection Check	Verify the **Gemini 1.5 Pro** API key is valid and the model is reachable.	API responds with a status code 200 and a valid JSON response.	Connection established.	Pass
PT-01	Response Time Test	Use a timer to measure the duration from clicking "Translate" to seeing the output.	The translation should be rendered in **under 3 seconds** for standard text.	2.4 seconds average.	Pass
PT-02	Concurrent API Load	Send multiple simultaneous translation requests from different sessions.	The **Gemini API** should process requests without significant latency or timeouts.	No significant lag.	Pass
PT-03	File Upload Load Test	Upload multiple heavy research **PDFs** simultaneously for academic translation.	The **Streamlit** app should process the files smoothly without crashing the server.	Handled 5 files at once.	Pass